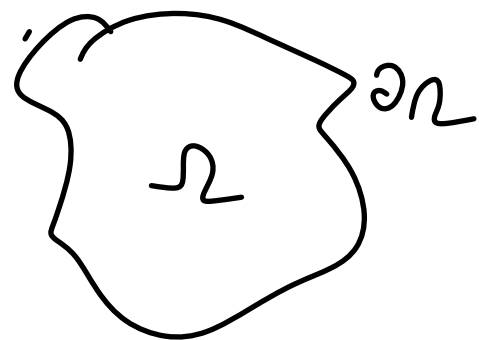


Interior Neumann problem

$$\Delta u = 0 \quad \text{in } \Omega$$

$$\frac{\partial u}{\partial n} = f \quad \text{on } \partial\Omega$$



Q: Is there always a solution?
 A: No, solution only exists as long as

$$\int_{\partial\Omega} f = \int_{\partial\Omega} \frac{\partial u}{\partial n} = \int_{\Omega} \Delta u = 0$$

Moreover, solution only determined upto a constant

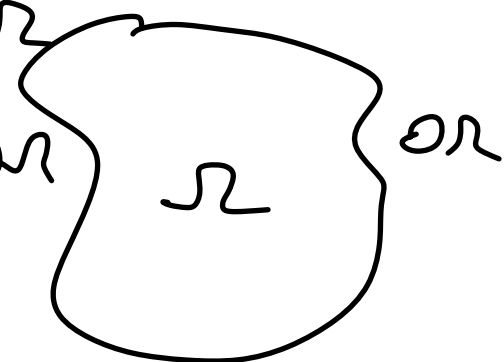
Fix: choose some normalization

Integral representation:

$$u = S[\sigma](x)$$

$\Delta u = 0$ in Ω

$\frac{\partial u}{\partial n} = f$ on $\partial\Omega$



$$\lim_{h \rightarrow 0^+} \nabla u(x_0 - h n(x_0)) \cdot n(x_0)$$

$$= \frac{1}{2} \sigma(x_0) - p \cdot v \cdot \frac{1}{2\pi} \int \frac{(x_0 - y) \cdot n(x_0) \sigma(y) dy}{|x_0 - y|^2} \leftarrow S'[\sigma](x_0)$$

$$= f(x_0)$$

$$\left(\frac{I}{2} + S' \right) \sigma = f$$

on $\partial\Omega$

HW: Show that

$$\int_{\partial\Omega} u(x) S'[\sigma](x) dx = \int_{\partial\Omega} \sigma(x) D^p u(x) dx$$

on

i.e. S' and D are adjoints of one another

Note
different
sign of
identity

Integral eqⁿ is second
kind, so are we done?

Unfortunately no! If σ is a constant on $\partial\Omega$

$$\text{then } \frac{1}{2} \sigma + D^p \sigma = 0 \text{ on } \partial\Omega$$

Recall $N(A) \subseteq N(A^*)$

so $\left(\frac{I}{2} + S' \right)$ also has a null vector

$$\left(\frac{\mathbf{I}}{2} + \mathbf{S}'\right) \sigma = f$$

$$\forall \sigma \quad \int \mathbf{1} \left(\frac{\mathbf{I}}{2} + \mathbf{S}'\right) \sigma = \int \sigma \left(\frac{\mathbf{I}}{2} + \mathbf{D}^{PV}\right) [\mathbf{1}] = 0$$

But $\int f \neq 0$ required for existence of solⁿs to PDE

So as long as data is consistent we are fine

How do we fix the null space? Add something that's not in the range!

$$\left(\frac{\mathbf{I}}{2} + \mathbf{S}'\right) \sigma + \frac{1}{|\partial \Omega|} \int_{\partial \Omega} \sigma \, ds_n = f \quad \text{is an invertible SKIE for } \sigma$$

$$\int \sigma = \int \left(\frac{\mathbf{I}}{2} + \mathbf{S}'\right) \sigma + \int \frac{1}{|\partial \Omega|} \int_{\partial \Omega} \sigma = \int f = 0$$

$$\Rightarrow \int_{\partial \Omega} \sigma = 0$$

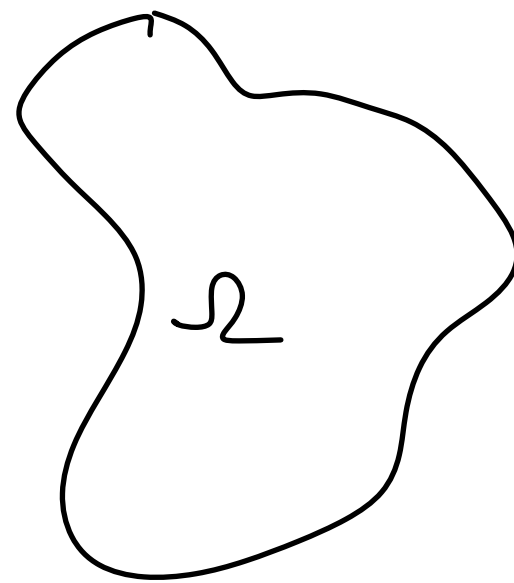
So if σ solves $\left(\frac{\mathbf{I}}{2} + \mathbf{S}'\right) \sigma = f$ it also solves $\int \sigma = 0$

Details on building a numerical test:

Consider INP again

$$\Delta u = 0 \quad \text{in } \Omega$$

$$\frac{\partial u}{\partial n} = f \quad \text{in } \partial\Omega$$



Step 1: Construct a known solⁿ to Laplace's equation in Ω

Say $u_{ex}(x) = G(x, x^*)$

Suppose now that $f(x) = \frac{\partial u^{ex}}{\partial n}(x, x^*) = \frac{\partial G}{\partial n}(x, x^*)$
for $x \in \partial\Omega$

then any solution to INP with this data must satisfy

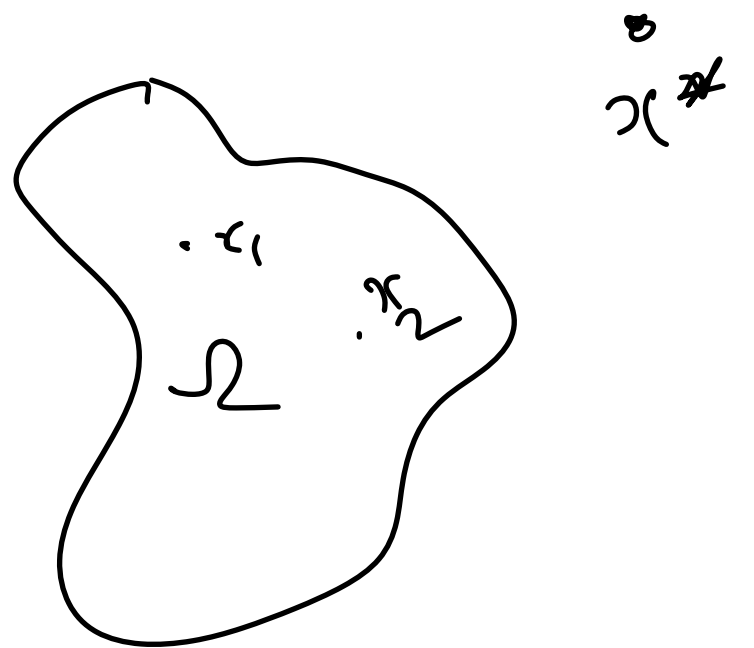
$$u(x) = u_{ex}(x) + c \quad \text{for a fixed constant } c$$

Analytic solution test :

Consider INP again

$$\Delta u = 0 \quad \text{in } \Omega$$

$$\frac{\partial u}{\partial n} = f \quad \text{in } \partial\Omega$$



Step 2: Solve INP using
our integral equation, i.e.

$$u(x) = S[\sigma](x) \text{ and}$$

σ solves

$$\frac{1}{2} \sigma(x) + S'[\sigma](x) = \frac{1}{|\partial\Omega|} \int_{\partial\Omega} \sigma = f(x) \quad \text{on } \partial\Omega$$

Step 3: After computing σ by solving IE above

Evaluate $u(x_j) = S[\sigma](x_j) = \int_{\partial\Omega} G(x_j, y) \sigma(y) ds_y \quad j=1,2$

Step 4: Verify

$$u(x_1) - u(x_2) = u_{\text{ex}}(x_1) + C - (u_{\text{ex}}(x_2) + C) = u_{\text{ex}}(x_1) - u_{\text{ex}}(x_2)$$

Integral equation $\rightarrow$ linear system

$$\sigma(x) + \int_{\partial\Omega} K(x,y) \sigma(y) ds_y = f(y)$$

K smooth in both x, y .

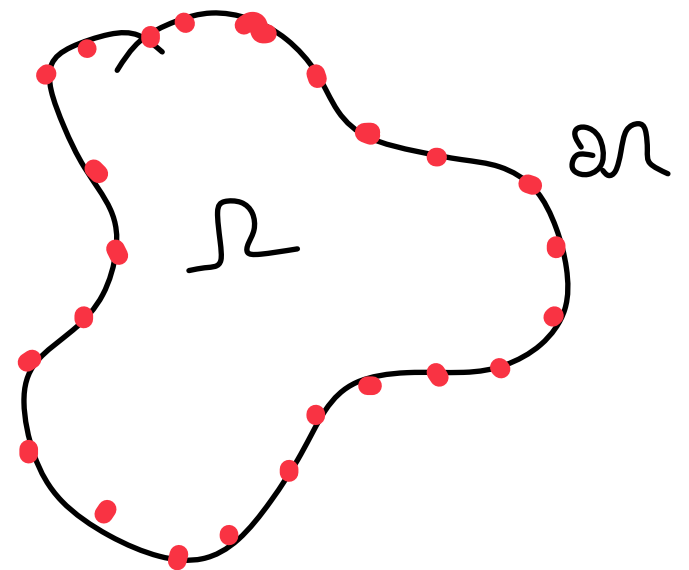
Suppose $\gamma(t) : [-\pi, \pi) \rightarrow \partial\Omega$
 $\gamma(t)$ periodic since curve is closed.

Let $x_j \in \mathbb{R}^2 = \gamma(t'_j)$ $t'_j = \frac{2\pi j}{2n+1}$

$\sigma_j = \sigma(x_j)$ $j = -n, \dots, n$
 $f_j = f(x_j)$

Discrete representation for σ, f of $\partial\Omega$

Also need ability to compute
 $\int K(x,y) \sigma(y)$



Quadrature rules

$$\int_{-\pi}^{\pi} f(t) \approx \sum f(t_j) w_j$$

(t_j, w_j)
 $\uparrow$ nodes
 $\nwarrow$ weights

Midpoint rule / periodic

trapezoid rule

$$\int_{-\pi}^{\pi} f(t) \approx \frac{2\pi}{2n+1} \sum_{j=-n}^n f(t_j)$$

is called a quadrature rule

$$t_j = \frac{2\pi j}{2n+1}$$

$$w_j = \frac{2\pi}{2n+1}$$

Q: what is the error?

$$f(t) = \sum_{m=-\infty}^{\infty} \hat{f}_m e^{imt}$$

$$\int_{-\pi}^{\pi} e^{imt} dt = 0$$

$$= 2\pi$$

if $m \neq 0$

if $m = 0$

$$\frac{2\pi}{2n+1} \sum_{j=-n}^n e^{imt_j}$$

$$= \frac{2\pi}{2n+1} \sum_{j=-n}^n e^{i m \frac{2\pi j}{2n+1}}$$

$= 1$ if $m \bmod 2n+1 = 0$
 $= 0$ otherwise

Error analysis periodic trapezoid rule

$$\int_{-\pi}^{\pi} f(t) dt = \int_{-\pi}^{\pi} \sum_{m=-\infty}^{\infty} \hat{f}_m e^{imt} dt = \hat{f}_0$$

$$\frac{2\pi}{2n+1} \sum_{j=-n}^n f(t_j) = \frac{2\pi}{2n+1} \sum_{j=-n}^n \sum_{m=-\infty}^{\infty} \hat{f}_m e^{im t_j}$$

$$= \hat{f}_0 + \sum_{m \neq 0} \hat{f}_{(2n+1)m}$$

$$\therefore \left| \int_{-\pi}^{\pi} f(t) dt - \frac{2\pi}{2n+1} \sum_{j=-n}^n f(t_j) \right| = \left| \sum_{\substack{m=-\infty \\ m \neq 0}}^{\infty} \hat{f}_{(2n+1)m} \right|$$

$$= O(n^{-(k-1)}) \text{ for } C^k \text{ functions}$$

$$= O(e^{-an}) \text{ for analytic functions}$$

Much better than
standard error
estimate $O(h^2)$
Need periodicity + smoothness
for better rate

$$\sigma(x) + \int_{\partial\Omega} K(x, y) \sigma(y) ds_y = f(x) \quad x \in \partial\Omega$$

$$\sigma(x) + \int_{-\pi}^{\pi} K(x, \gamma(t)) \sigma(\gamma(t)) |\gamma'(t)| dt = f(x) \quad x \in \partial\Omega$$

Nyström method: Replace integral by quadrature rule & enforce equation at quadrature nodes

$$\sigma(x) + \sum_{j=-n}^n K(x, \gamma(t_j)) |\gamma'(t_j)| w_j \sigma_j = f(x) \quad x \in \partial\Omega$$

$$x = \gamma(t_m) = x_m$$

$$\sigma_m + \sum_{j=-n}^n K(\gamma(t_m), \gamma(t_j)) |\gamma'(t_j)| w_j \sigma_j = f_m$$

$$A_{m,j} = \delta_{m,j} + K(\gamma(t_m), \gamma(t_j)) |\gamma'(t_j)| w_j$$

Nice property of $\mathcal{SKE}$ After computing σ_m , σ on rest of domain given by

$$\sigma(x) = f(x) - \int K(x, y) \sigma(y) dy$$

$$\sigma(x) = f(x) - \sum_{j=-n}^n K(x, \tau(t_j)) \sigma_j |\tau'(t_j)| w_j$$

Laplace kernels

D: Double layer potential $\frac{(x-y) \cdot n(y)}{2\pi |x-y|^2} = K_D$

S': Normal derivative of single layer $-\frac{(x-y) \cdot n(x)}{2\pi |x-y|^2} = K_{S'}$

$$\lim_{x \rightarrow y} K_D(x, y) = \frac{K(x)}{2\pi}$$

PTR demo

Other kernels

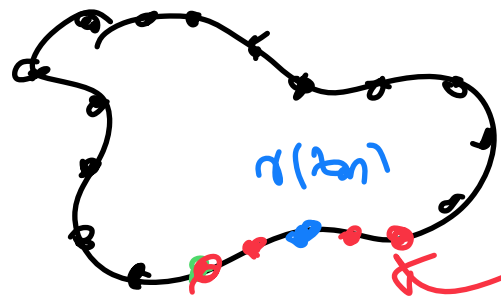
Helmholtz double layer potential $(\Delta + k^2)u = 0$

$$K_{Dn} = \frac{i}{4} \frac{\partial}{\partial n} H_0^{(1)}(k|x-y|)$$

as $x \rightarrow y$

$$K_{Dn}(x, y) = O(\log|x-y|)$$

Singular on diagonal



update weights

Many options available

$$\sum_{j=-n}^n K(r(b_m), r(b_j)) \sigma_j w_j$$

$$\rightarrow \sum_{m, j \text{ close}} K(r(b_m), r(b_j)) \sigma_j w_{jm}$$

$\sigma_j w_{jm}$

$$+ \sum_{m, j \text{ far}}$$

$$K(r(b_m), r(b_j)) \sigma_j w_j$$

(Fast algorithm comparable!)

Product quadrature rules (Kress, Helsing)

Quadrature by expansion

Singularity subtraction

Locally corrected GGO based rules

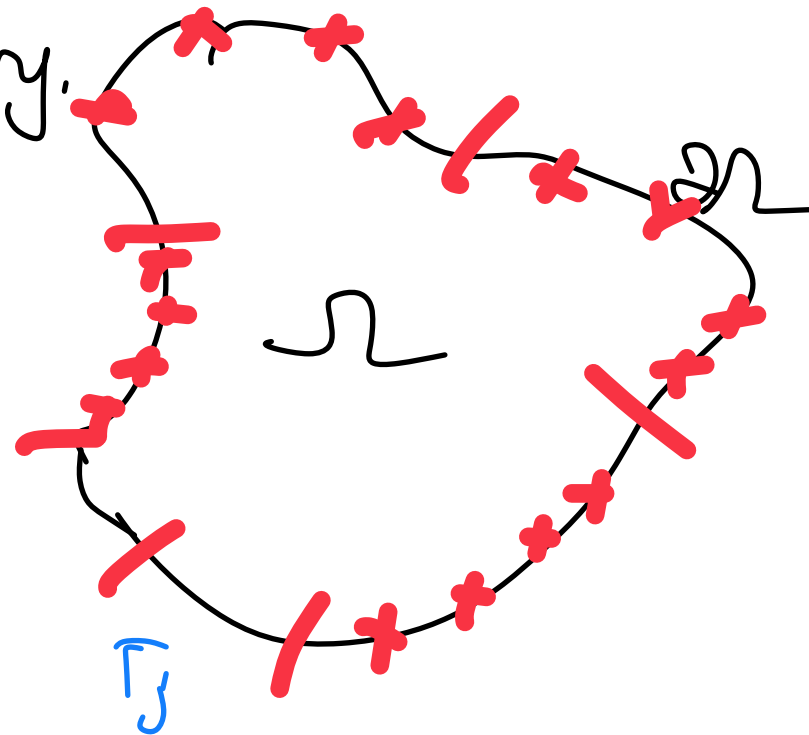
How to convert integral equation into linear system

Step 1: Discretize the boundary.

Suppose, we are given

$\gamma(t) : [0, 1] \rightarrow \mathbb{R}^2$, a
parametrization for $\partial\Omega$

$$\gamma(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$$



Use adaptive chebyshev discretization discussed in lec 1/2 to resolve $\begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$ $\begin{bmatrix} x'(t) \\ y'(t) \end{bmatrix}$ & so on

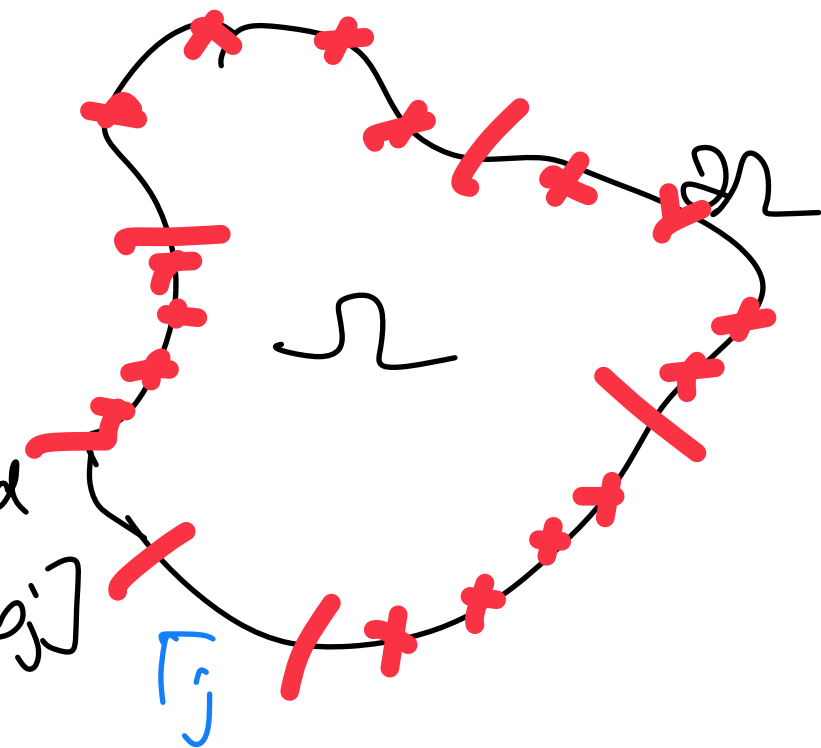
$\Gamma_j = \gamma[a_j, b_j]$ be a
collection of panels on
the boundary

where $\cup_j [a_j, b_j] = [0, 1]$
& $(a_j, b_j) \cap (a_k, b_k) = \emptyset$ for $j \neq k$



On each panel Γ_j
 let $[x_{j,l}] = [y_{j,l}] = [x(t_{j,l})]$ be

the Chebyshev/Legendre nodes
 on the panels where $t_{j,l}$ correspond
 to Chebyshev/Legendre nodes on $[a_j, b_j]$



Integral equation of
 interest

$$\sigma(x) + \int_{\partial\Omega} K(x,y) \sigma(y) ds_y = f(x)$$

Recall $\cos(\frac{2\pi j}{2n+1})$ Chebyshev
 nodes on $[-1,1]$

$$\text{then } \frac{a+b}{2} + \frac{(b-a)}{2} \cos(\frac{2\pi j}{2n+1})$$

are Chebyshev nodes on
 $[a,b]$

Suppose for now $K(x,y)$ is a
 smooth function of both x and y .

Nystrom method is one where we will
 approximate σ, f by their samples $\sigma_{j,l} = \sigma\left(\begin{bmatrix} x_{j,l} \\ y_{j,l} \end{bmatrix}\right)$

$$\sigma(x) + \int_{\Omega} K(x,y) \sigma(y) dy = f(x)$$

Need the ability to compute integrals like the one above.

Dehor: how to compute $\int_1^1 g(t) dt$?

Interpolatory quadrature:

Replace g by its interpolant $\tilde{g}(t)$
 & use integrals of basis functions
 (typically know analytically)

$$\tilde{g}(t) = \sum_{n=0}^m c_n \phi_n(t)$$

$$c_m = A \begin{bmatrix} g(t_0) \\ \vdots \\ g(t_m) \end{bmatrix}$$

$$\text{Let } \int \phi_n(t) = \alpha_n$$

$$\text{then } \int_1^1 g(t) dt \approx \int_1^1 \tilde{g}(t) dt = \sum_{n=0}^m c_n \alpha_n = \sum_{j=0}^m g(t_j) w_j$$

$$\int_{-1}^1 g(t) dt \approx \int_{-1}^1 \tilde{g}(t) dt = \sum_{j=0}^n g(t_j) w_j$$

quadrature weights

quadrature nodes

For polynomial interpolant, since

$$|g - \tilde{g}|_1 \approx O(n^{-(k-1)}) \text{ for } C^k \text{ functions}$$

error

$$\left| \int_{-1}^1 (g(t) - \tilde{g}(t)) dt \right| \approx O(n^{-(k-1)})$$

for C^k functions

If function represented with p -Chebyshev nodes & N panels (each of size h). Then as we decrease the size of the panels both interpolation & quadrature error decrease like $O(h^p)$ $k \geq p$